Math 115A Spring 2026 Discussion 11 Handout

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1 Introduction

Today, we'll say more about linear transformations - we'll describe some properties of isomorphisms of vector spaces, then talk about representing linear transformations using matrices.

2 Vector Spaces Isomorphisms

The basic definition of a vector space isomorphism is:

Definition: Let V and W be vector spaces over a field $\mathbb{F}$. A linear transformation $f : V \to W$ is called a vector space isomorphism if there is another linear transformation $g : W \to V$ such that the compositions $f \circ g$ and $g \circ f$ are both the identity map. (That is, if f has an inverse).

Exercise: What are the domain and codomain for $f \circ g$? What are the domain and codomain for $g \circ f$?

In practice, we rarely use this definition. For one thing, as you've seen in lecture:

Theorem: A linear transformation $f : V \to W$ is an isomorphism if and only if it is bijective.

Exercise: If you don't remember how to prove this, theorem, prove it.

Here's another useful characterization:

Theorem: Let $f : V \to W$ be a linear transformation and $B = \{v_1, \ldots, v_n\}$ a basis for V. The map f is an isomorphism if and only if $f(B) := \{f(v_1), \ldots, f(v_n)\}$ is a basis for W.

Exercise: Prove it.

Exercise: As an example, consider the map $f : \mathbb{R}^2 \to \mathbb{C}$ that sends (a, b) to $2(a + b) - (3b)i$. Show that f is an isomorphism of $\mathbb{R}$-vector spaces.

Exercise: On the other hand, show (using any definition you want) that the map $g : \mathbb{R}^2 \to \mathbb{R}$ given by $g(a, b) = 2a + 3b$ is not an isomorphism.

Exercise: Let $V = \mathbb{R}^3$. Let $V^* = \mathcal{L}(V, \mathbb{R})$ be the vector space of linear transformations $\mathbb{R}^3 \to \mathbb{R}$. (We call V^* the *dual* of V). Find an isomorphism $V \to V^*$. [Hint: Fix a basis B of V, and try to find a "corresponding" basis for V^*. As a hint, remember that every function in V^* is defined by what it does to the elements of B].

Everything above implies that two vector spaces (over a given field $\mathbb{F}$) are isomorphic if and only if they have the same dimension; but note that not every map between isomorphic vector spaces is an isomorphism.

Exercise: Give an example of a linear transformation $f : \mathbb{R}^3 \to \mathbb{R}^3$ which is not an isomorphism.

That being said, once you know that two (finite-dimensional) vector spaces have the same dimension, you can show that a map between them is an isomorphism with less work.

Exercise: Let V and W be finite-dimensional vector spaces over a field $\mathbb{F}$ such that $\dim(V) = \dim(W)$. Let $f : V \to W$ be a linear transformation.

(a) Show that if f is injective, then f is an isomorphism.

(b) Show that if f is surjective, then f is an isomorphism.

Challenge: Show that this is not true if the vector spaces are not finite dimensional. [Hint: you've seen a counterexample before!]

Hint: Consider the effect of f on a basis of V.

When the dimension of V and W is large, it can be difficult to determine wither a linear map $f; V \to W$ is an isomorphism; the above result makes things a bit easier - rather than showing both injectivity *and* surjectivity, it's enough to show one or the other.

In lecture, the following theorem was shown:

Theorem: Let V, W, V' be vector spaces over a field $\mathbb{F}$, such that $V \cong W$. Then $\mathcal{L}(V, V') \cong \mathcal{L}(W, V')$ and $\mathcal{L}(V', V) \cong \mathcal{L}(V', W)$. (Here, $\mathcal{L}(V, V')$ is the vector space of linear maps $V \to V'$, etc.).

If you assume the spaces are finite dimensional, you can prove this by comparing the dimensions of the various spaces. But the statement is still true when the vector spaces are infinite-dimensional; to prove it, we can fix an isomorphism $f : V \to W$, and build associated isomorphism between the spaces of linear transformations. I can go over this in more detail in discussion if there's demand.

3 Matrices

As some of the examples in the previous section show, it is often easier to understand a linear transformation $f : V \to W$ after choosing bases for V and W. (By the way, this suggests an easy way to "graph" a linear transformation $f : V \to W$: just sketch a basis for V and then indicate the image of this basis in W). Once these bases have been chosen, all the information one might need about a linear transformation can be expressed in a matrix.

This viewpoint can be quite overwhelming when you first see it, so it's helpful to get a bit of concrete practice with the procedure of multiplying a matrix by a vector; once you have a feel for how these computations go, it's much easier to handle abstract problems.

Exercise: Multiply:

(a)

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 4 & 4 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 7 \\ -1 \end{pmatrix}$$

(d)

$$\begin{pmatrix} 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 7 \\ 6 \\ 0 \end{pmatrix}$$

Exercise: What happens when we multiply the matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 4 & 4 & 2 \end{pmatrix}$$

By the "standard basis" elements $\left(\begin{smallmatrix} 1 \\ 0 \\ 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} 0 \\ 0 \\ 1 \end{smallmatrix}\right)$?

How can we apply this to abstract linear transformations? Well, we first need to know how to go from an abstract element v in an abstract vector space V to a column vector. In order to do so, we need to fix a basis. So, let's do that now: take $B = \{v_1, \ldots, v_n\}$ an *ordered* basis. Recall that we can express v (uniquely) as a linear combination

$$v = \sum_i a_i v_i$$

Then with respect to the basis B, we can write v as a column vector:

$$v = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

Again, notice that the order of the elements in the basis matters! Given an ordered basis $B' = \{w_1, \ldots, w_m\}$ for W, we can likewise express any element $w \in W$ uniquely as a sum $w = \sum_i b_i w_i$, and hence as a column vector

$$w = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Now, consider a linear transformation $T : V \to W$.

Exercise: Recall that $B = \{v_1, \ldots, v_n\}$ was our basis for V and $B' = \{w_1, \ldots, w_n\}$ was our basis for W. Suppose that for each $1 \leq j \leq n$, we have $Tv_j = \sum_{i=1}^m b_{ij} w_i$.

(a) How would you write Tv_1 as a column vector? What about Tv_2, etc.

(b) Using (a), write T as a matrix.

Here's a more concrete exercise, which shows the power of this approach:

Exercise: Consider the map $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ given by rotating the plane counterclockwise by an angle of θ.

(a) Convince yourself, geometrically, that R_θ is linear.

(b) Express R_θ as a matrix, with respect to the standard basis for $\mathbb{R}^2$. (Hint: this is hard to do if you try to write it down "in one gulp." So instead, consider what R_θ does on a basis of $\mathbb{R}^2$).

Exercise: Let $V = P_{\leq 3}(\mathbb{R})$, the space of polynomials of degree at most 3 with real coefficients. Let $D : V \to V$ be the map that sends a polynomial $p(x)$ to its derivative $p'(x)$.

(a) Choose a basis for V.

(b) With respect to the basis you picked in (a), express D as a matrix.

4 Change of Basis

Since there are many available choices of basis for a given vector space, there are many ways to express the same matrix:

Confusing Exercise: Consider the linear transformation $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ from before, and take the new "non-standard" basis $B' = \{\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)\}$ for $\mathbb{R}^2$; find a matrix representation for R_θ in terms of B'.

Since we're used to working with vectors "in coordinates," it can be difficult to wrap your head around how a change of basis works; I think this is the most confusing topic in linear algebra. When thinking about linear transformations in terms of matrices, it might be helpful to view such matrices M not just as maps between vector spaces, but as maps of *pairs* $M : (V, B) \to (W, B')$, where

B is a basis for V and B' is a basis for W. (Note: there is no requirement here that M "takes B to B';" what we mean is that the column vectors you "feed into" M should be written with respect to the basis B and the column vectors that M "spits out" should be written with respect to the basis B').

In particular, suppose we have a matrix M that gives a linear transformation $(V,B) \to (W,B')$; let $T : V \to W$ be the linear transformation M represents. Let C be a different basis for V. How can we represent T as a matrix $(V,C) \to (W,B')$?

The idea is as follows. Let $B = \{b_1, \ldots, b_n\}$ and $C = \{c_1, \ldots, c_n\}$. We want a composition of matrix-maps like so:

$$(V,C) \xrightarrow{S_{C,B}} (V,B) \xrightarrow{M} (W,B')$$

where $S_{C,B}$ is a matrix which takes a column vector representing a given vector v with respect to the basis C, and takes it to a column vector representing v with respect to the basis B. To write down this matrix, it's enough to write down each c_i as a column vector with respect to B. The map $(V,C) \to (W,B')$ is given by the matrix product $MS_{C,B}$.

Likewise, if we want to change the basis of W from B' to D, we should find the analogous matrix $S_{B',D} : (W,B') \to (W,D)$, and multiply on the left to get $S_{B',D}M$.

In particular, suppose we have a matrix-map $M : (V,B) \to (V,B)$ and want to rewrite it terms of a basis B'. We just find the matrices $S_{B,B'}$ and $S_{B',B}$ and compose on either side:

$$(V,B') \xrightarrow{S_{B',B}} (V,B) \xrightarrow{M} (V,B) \xrightarrow{S_{B,B'}} (V,B')$$

Exercise: Show that $S_{B,B'}^{-1} = S_{B',B}$.

Exercise: Using change of basis matrices, redo the "confusing exercise" at the beginning of the section.